

Executive Summary Report

1. Executive Overview

This report summarizes the end-to-end data analytics work completed across the ERP Data Analytics internship assignment: profiling and cleaning a 5,250-row denormalized ERP export, designing a normalized (3NF) SQL Server database, writing 23 analytical SQL queries, and building a Power BI executive dashboard. The objective throughout was to produce analysis that is fully traceable to real data — every figure in this report is derived directly from the cleaned dataset (02_Cleaned_Dataset.csv) and the SQL Server database (ERP_System), with no estimated, averaged, or fabricated values used to fill gaps in the source data.

Where the source data itself was incomplete, inconsistent, or structurally limited, this report documents that honestly rather than presenting a misleading number. Sections 5 and 6 below cover these limitations directly, since they are as important to the business picture as the figures that could be calculated cleanly.

2. Key Performance Indicators

The table below summarizes the core business metrics calculated from the full cleaned dataset (SalesOrders table, ERP_System database).

Metric	Value
Total Sales Revenue	Rs. 9,005,235,327 (≈ Rs. 9.01 billion)
Total Order Count	5,182
Total Customers (distinct records)	5,067
Average Order Value (AOV)	Rs. 4,130,842 (calculated on the ~42% of orders with complete Quantity/UnitPrice data)
Total Products (distinct records)	2,751
Return Rate	100% — a source data artifact, not real return behavior (see Section 5)

Note on AOV: Average Order Value is calculated only across orders with a non-null Quantity and UnitPrice (approximately 42% of all 5,182 orders). The remaining orders have genuinely missing

pricing data in the source file and were correctly excluded from this average rather than estimated — see the Data Profiling Report (01_Data_Profiling_Report) for full detail on this null-handling approach.

3. Sales Performance Insight

3.1 Monthly Revenue Trend

Revenue was tracked monthly from July 2024 through July 2026 (the full date range present in the source data, excluding 1,690 rows flagged as invalid future dates). Revenue shows a general upward trend through 2024 and 2025, followed by a partial 2026 (only 7 months of data are available for 2026, since this is the most recent month present in the source file). A calendar-year-over-year comparison would misleadingly suggest a steep decline in 2026 simply because fewer months of data exist for that year; a fair like-for-like comparison (January–July only, across all three years) gives a more accurate read on real year-over-year performance and is the version used for trend analysis in this report.

3.2 Top 10 Customers by Spend

The single largest entry in the Top 10 Customers ranking is an aggregated “Unknown” group (Rs. 194,979,324, Lahore/PK) rather than one real individual customer. This reflects the ~22% of CustomerName values that were missing in the source data and filled with the placeholder “Unknown” during cleaning; because all such rows share the identical name, they are grouped together by any customer-level aggregation, inflating this single bucket relative to any one real named customer. The remaining 9 positions in the Top 10 are genuine, individually named customers.

Rank	Customer	City / Country	Total Spend
1	Unknown (aggregated)	Lahore / PK	Rs. 194,979,324
2–10	9 individually named customers	Various	Rs. 40M – 65M each

3.3 Sales Breakdown by Product Category

All seven product categories perform within a relatively narrow band of one another (Rs. 985 million – Rs. 1.44 billion), with Medical leading and Food lowest. This even spread suggests a diversified revenue base rather than dependence on any single category.

Category	Revenue	Order Count
Medical	Rs. 1,439,707,363	723
Office	Rs. 1,376,663,394	747
IT	Rs. 1,359,298,925	677
Fashion	Rs. 1,221,142,100	713
Furniture	Rs. 1,220,774,469	732
Electronics	Rs. 1,219,210,656	737
Food	Rs. 984,776,483	696

3.4 Sales Breakdown by Country

Pakistan accounts for the largest share of both revenue and order volume, consistent with it having the most orders overall, followed by UAE and Saudi Arabia.

Country	Revenue	Order Count
Pakistan (PK)	Rs. 3,029,664,420	1,992
UAE (AE)	Rs. 2,647,084,106	1,513
Saudi Arabia (SA)	Rs. 1,989,268,531	1,002

A cross-check of the City and Country fields during Task 1 profiling found no consistent geographic relationship between them (each city appears in near-equal proportion across all three countries) — this is documented in the Data Profiling Report as a data quality limitation, not corrected by altering the Country field, since doing so would contradict the assignment's explicit ISO-code standardization requirement and this exact required Country breakdown.

4. Operational Insights

4.1 Order Fulfillment & Payment Status

Shipment status (Completed, Processing, Cancelled, Pending, Unknown) and Payment status show a near-uniform distribution across categories, each within a few percentage points of 24–25% (with Unknown at ~3%). This close-to-random split suggests these status fields may not encode a genuine operational pattern in the underlying data, and should not be read as evidence of an operational bottleneck or strength in any one status.

4.2 Payment Method Distribution

All five payment methods (Bank Transfer, Online, Cheque, Cash, Card) are used in near-equal proportion (each between 19% and 20% of transactions), again suggesting no strong customer preference signal is present in this data, likely for the same structural reason as the status fields above.

4.3 Supplier Performance

Supplier performance was assessed on total purchase value and product diversity, since a true on-time fulfillment rate could not be calculated — no promised or actual delivery date field exists anywhere in the source data. Watts, Mahoney and McKay led named suppliers by total purchase value; as with customers and products, an aggregated “Unknown” supplier group leads by sheer row count due to missing SupplierName values.

4.4 Employee Performance

Michael Singh led named employees by revenue handled (Rs. 65.3 million). As with customers/suppliers, an aggregated “Unknown” employee group leads overall due to missing EmployeeName values in the source data.

5. Data Limitations Affecting Analysis

Three areas of the required analysis are directly affected by structural limitations in the source dataset. These are reported transparently here rather than papered over, consistent with the data-handling approach used throughout this project.

- **Return Rate (100%):** The source data assigns a ReturnID to nearly every row (~97%) regardless of whether a product return actually occurred — there is no field distinguishing a genuine return from a populated placeholder ID. The resulting 100% figure is a data artifact, not a real indicator that every order was returned, and should not be acted on as a literal business metric.
- **Supplier On-Time Fulfillment Rate:** Not computable. No promised or actual delivery date exists anywhere in the source data (only OrderDate is present for purchase orders), so this metric was omitted rather than approximated from an unrelated field.
- **Inventory Valuation & Low-Stock Alerts:** Not computable. No stock-on-hand quantity or reorder-point threshold exists in the source data. A proxy calculation (Purchased minus Sold, per product) was tested but found to be structurally meaningless: because the source file is a single flat table combining multiple business events per row, Purchase quantity and Sales quantity for a given product are drawn from the same underlying row, so the

subtraction returns approximately zero for every product rather than a real stock balance. This metric was therefore omitted.

Full detail on all data quality findings — including referential integrity anomalies, placeholder/junk data, and the complete before/after cleaning audit — is documented in 01_Data_Profiling_Report.

6. Methodology Notes

- Total Sales Amount was calculated as: $\text{Quantity} \times \text{UnitPrice} \times (1 - \text{Discount}/100)$, applied only to rows with non-null Quantity and UnitPrice. No missing value was estimated or averaged to compute this figure.
- All “Unknown” groupings in customer, supplier, and employee breakdowns represent multiple distinct real-world entities whose identifying name was missing in the source data — not a single large entity.
- Database design (Task 3) used newly generated surrogate keys for every table rather than the raw source IDs, since Task 1 profiling found source ID columns (CustomerID, SupplierID, ProductID, EmployeeID, and even SalesOrderID) did not map reliably to a single real-world entity.
- All SQL analytics queries exclude invalid future-dated orders (DateFlag = 'Invalid - Future Date') from trend-based calculations (Monthly Revenue Trend, YoY Growth) to avoid distorting business trend analysis with known-bad dates; these rows remain visible in the underlying tables for audit purposes.

7. Conclusion

This project delivered a complete, auditable pipeline from a messy 27-column ERP export to a normalized SQL Server database, verified analytics, and an executive Power BI dashboard. Every figure in this report and the accompanying dashboard is traceable to the cleaned dataset and documented cleaning decisions, and every area where the source data could not honestly support a required metric has been disclosed rather than filled in. This approach — flagging rather than fabricating, and documenting limitations alongside findings — was maintained consistently across all five tasks of this assignment.